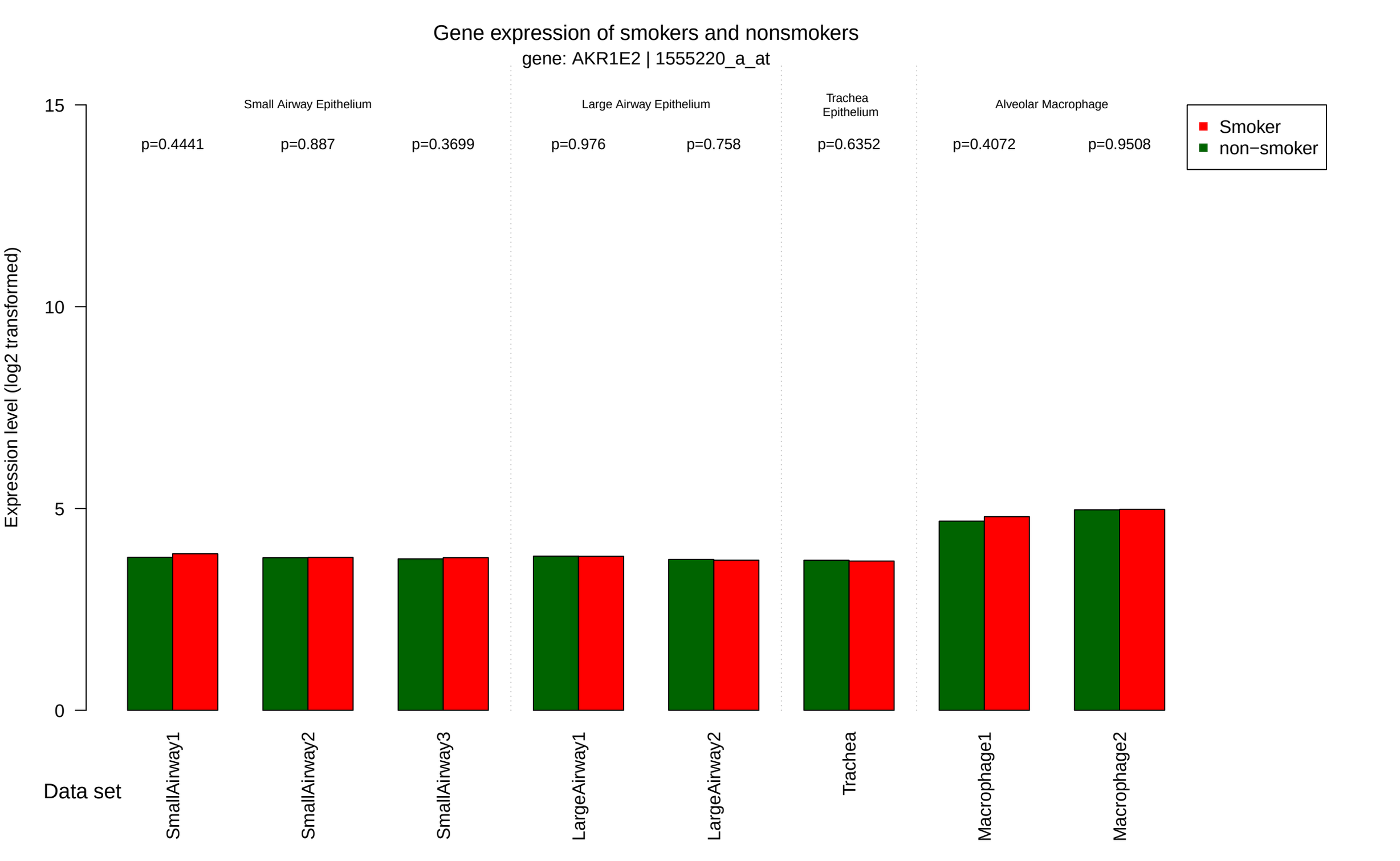
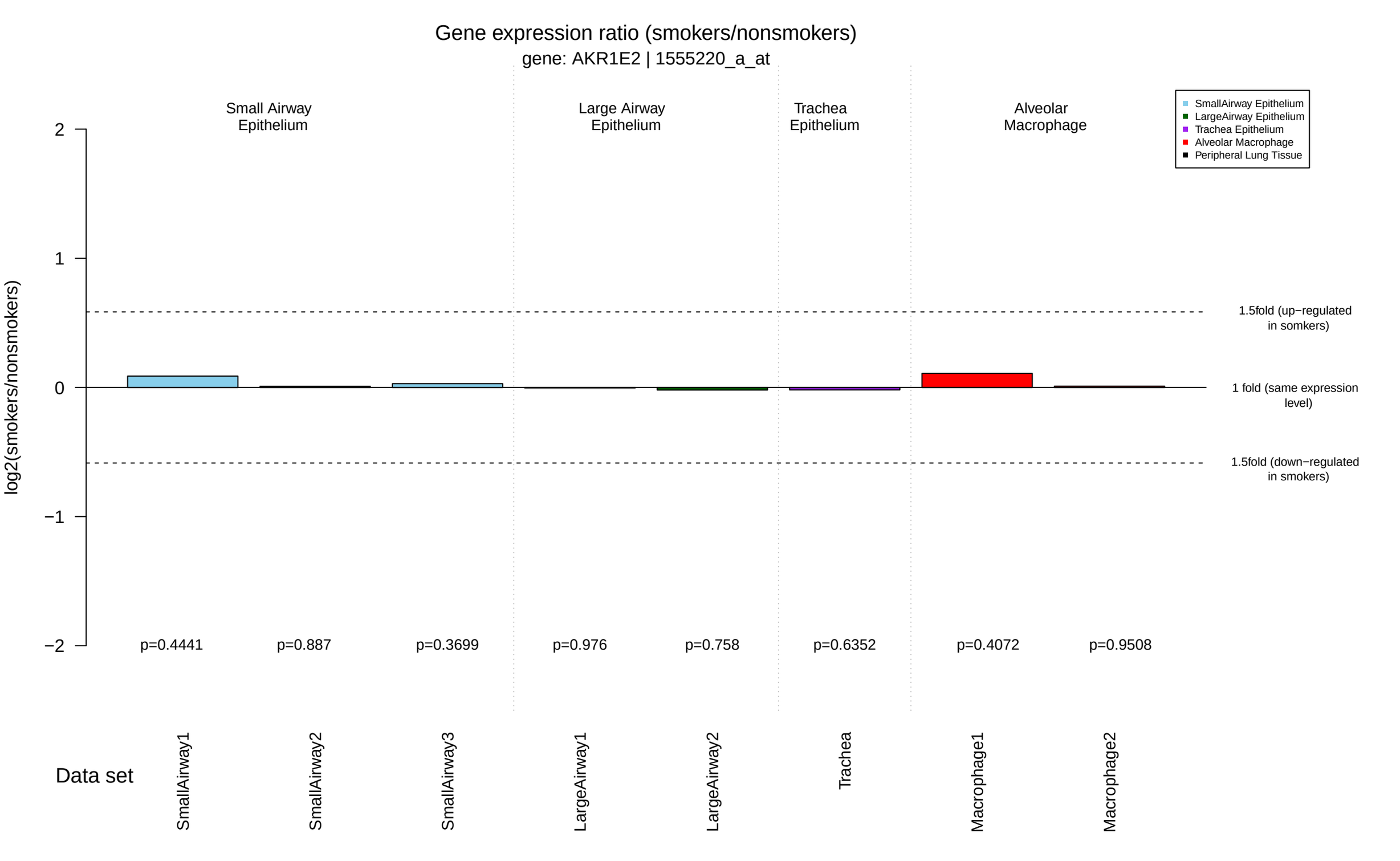
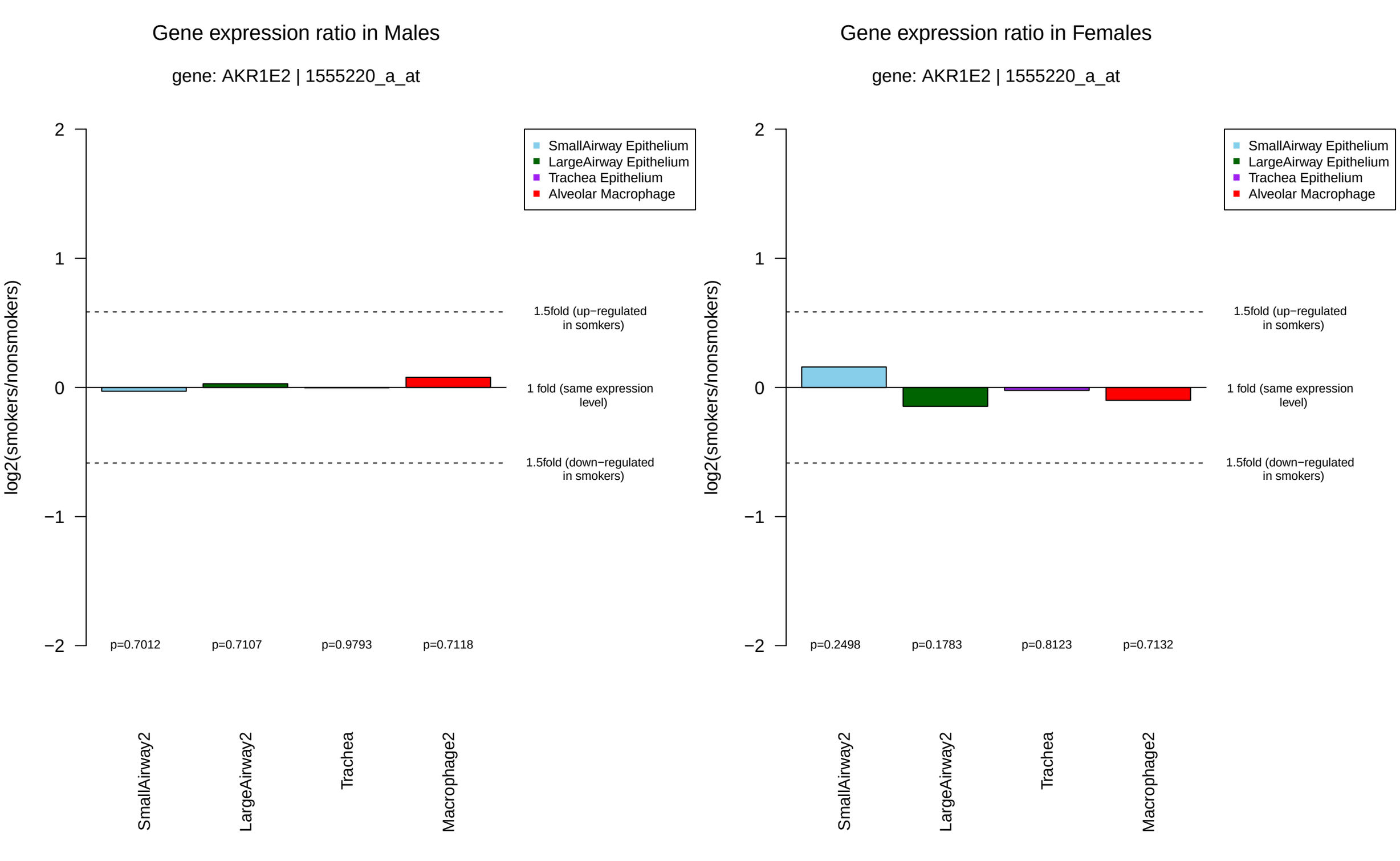
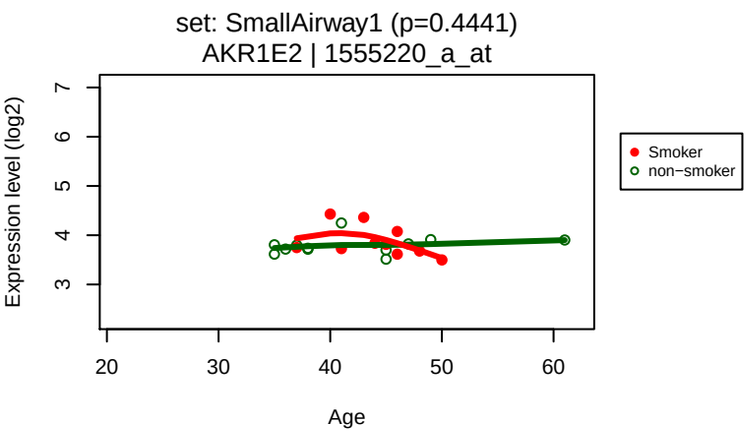
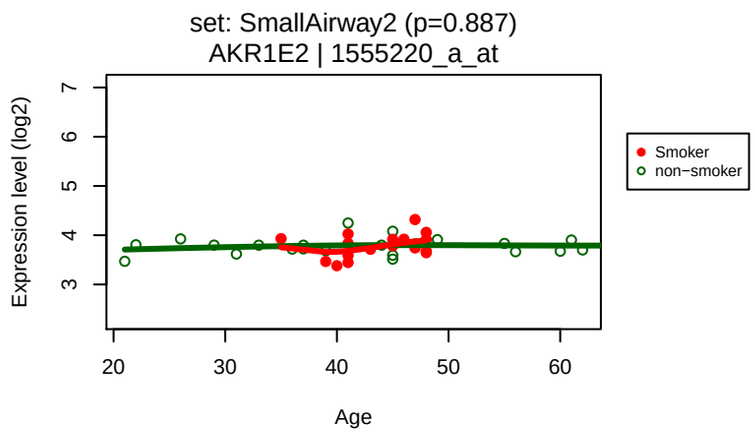
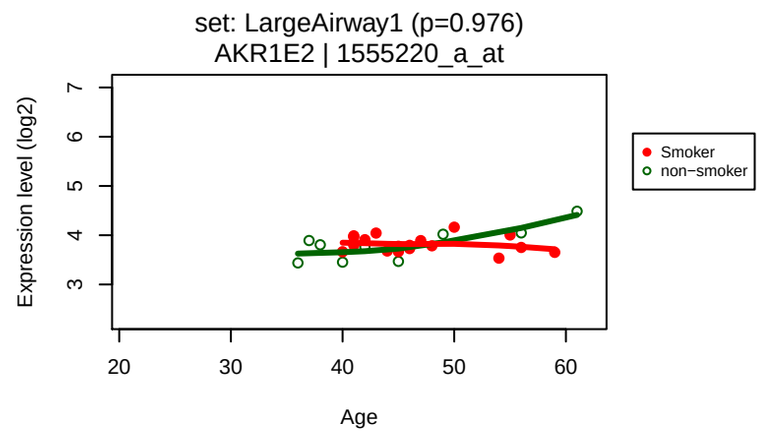

Gene expression of smokers and nonsmokers
gene: AKR1E2 | 1555220_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4441
p=0.887
p=0.3699
p=0.976
p=0.758
p=0.6352
p=0.4072
p=0.9508
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

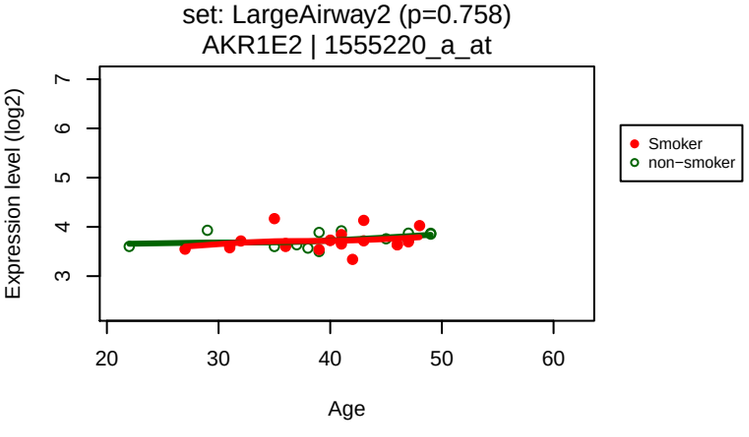

Gene expression ratio (smokers/nonsmokers)
gene: AKR1E2 | 1555220_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4441
p=0.887
p=0.3699
p=0.976
p=0.758
p=0.6352
p=0.4072
p=0.9508
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

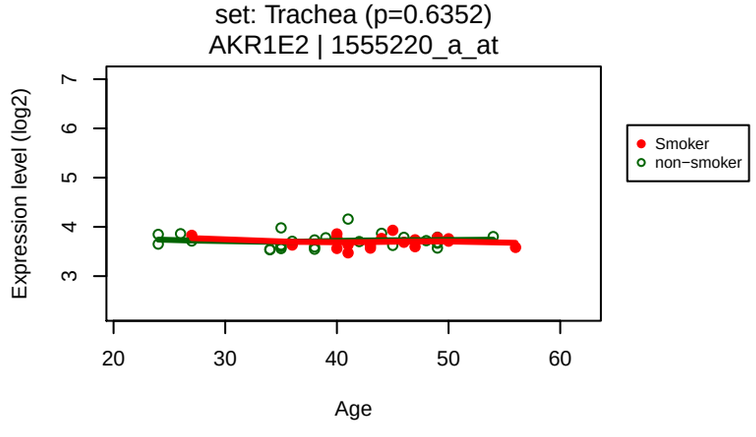

Gene expression ratio in Males
gene: AKR1E2 | 1555220_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7012
p=0.7107
p=0.9793
p=0.7118
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKR1E2 | 1555220_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2498
p=0.1783
p=0.8123
p=0.7132
SmallAirway2
LargeAirway2
Trachea
Macrophage2

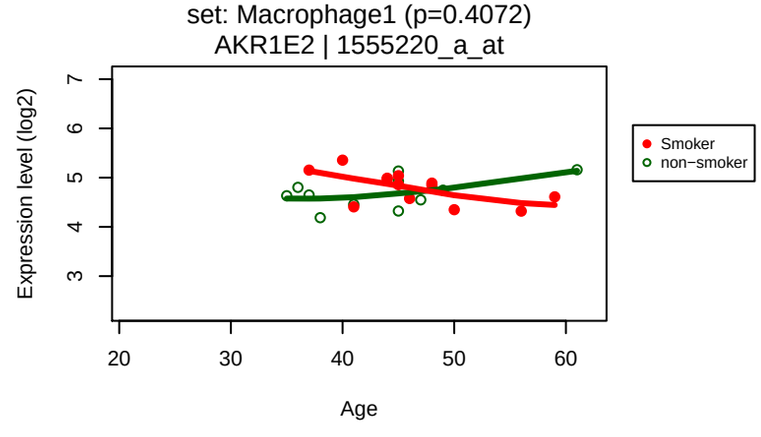

set: SmallAirway1 (p=0.4441)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker

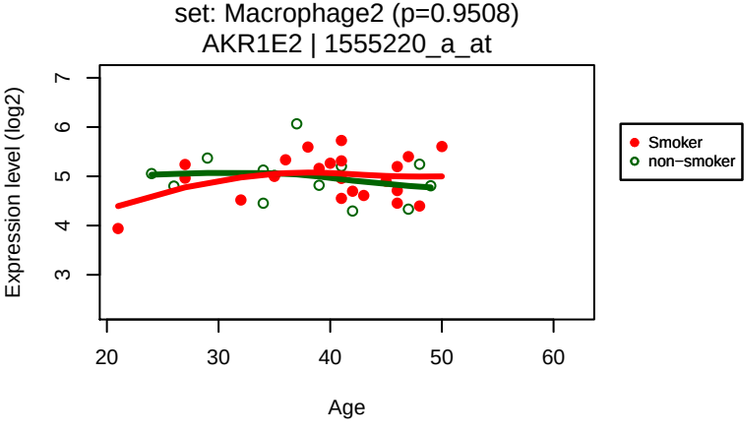

set: SmallAirway2 (p=0.887)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker

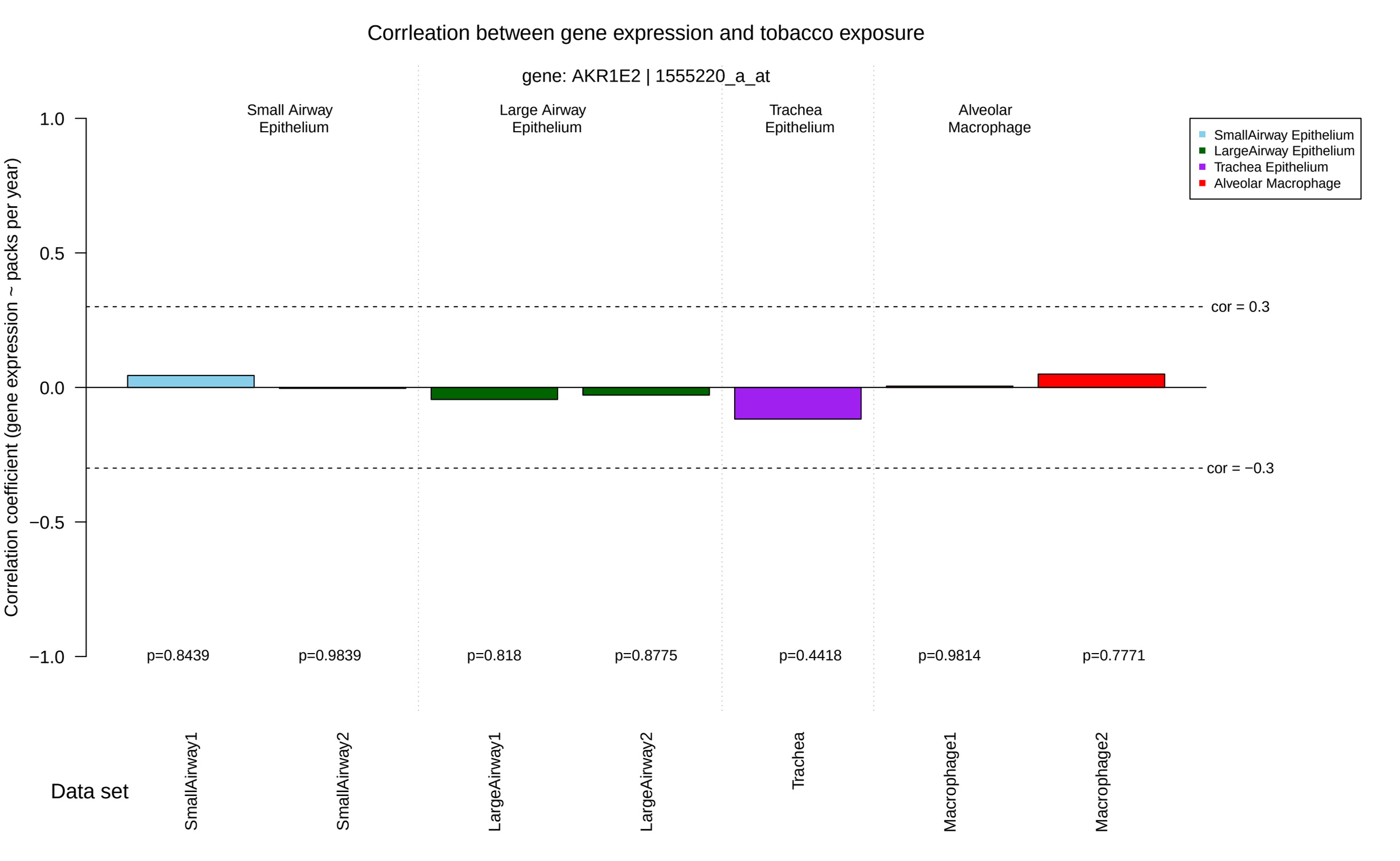

set: LargeAirway1 (p=0.976)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.758)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6352)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4072)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9508)
AKR1E2 | 1555220_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKR1E2 | 1555220_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8439
p=0.9839
p=0.818
p=0.8775
p=0.4418
p=0.9814
p=0.7771
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2